

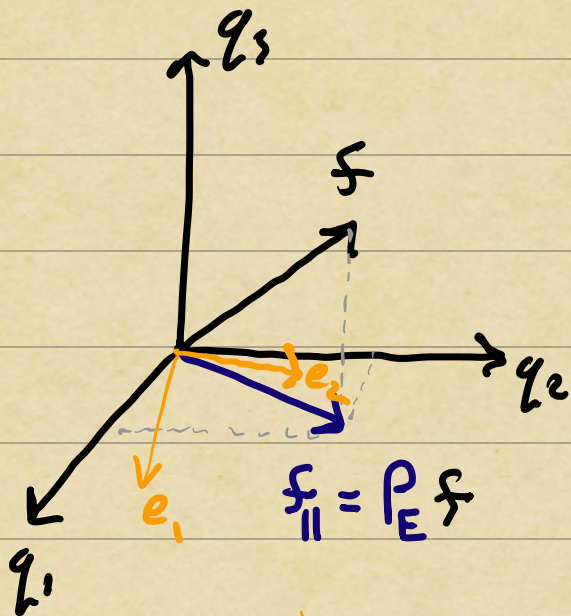
What Makes a "Good" Basis?

Goal: Minimize $\|f - E c\|_H$, where $f \in H$,
 $c, e_1, \dots, e_n$

$$E = \begin{bmatrix} | & & | \\ e_1 & \dots & e_n \\ | & & | \end{bmatrix} : \mathbb{R}^n \rightarrow H$$

If $E^* E = I$, then $c = E^* f$.

$$\begin{bmatrix} \langle e_1, e_1 \rangle & \dots & \langle e_n, e_1 \rangle \\ \vdots & & \vdots \\ \langle e_1, e_n \rangle & \dots & \langle e_n, e_n \rangle \end{bmatrix} \begin{bmatrix} \langle f, e_1 \rangle \\ \vdots \\ \langle f, e_n \rangle \end{bmatrix}$$



Two Step Procedure:

$$\Rightarrow \text{Project } f_{||} = P_E f$$

$$\Rightarrow \text{Solve } E c = f_{||}$$

$$E = \begin{bmatrix} | & | \\ e_1 & e_2 \\ | & | \end{bmatrix}$$

$$\begin{aligned} P_E f &= \langle f, q_1 \rangle q_1 + \langle f, q_2 \rangle q_2 \\ &= \underbrace{\begin{bmatrix} | & | \\ q_1 & q_2 \\ | & | \end{bmatrix}}_Q \begin{bmatrix} \langle f, q_1 \rangle \\ \langle f, q_2 \rangle \end{bmatrix} \\ &= Q Q^* f = f_{||} \end{aligned}$$

Key Idea: QR Decomp.

$$\begin{matrix} E & Q & R \\ \begin{bmatrix} | & | \\ e_1 & \dots & e_n \\ | & & | \end{bmatrix} & \begin{bmatrix} | & | \\ q_1 & \dots & q_n \\ | & & | \end{bmatrix} & \begin{bmatrix} x & x & \dots & x \\ & x & & x \\ & & \ddots & \\ & & & x \end{bmatrix} \end{matrix}$$

$$E \underline{e} = \underline{f}_{||} \Rightarrow (QR) \underline{e} = \underline{f}_{||} = QQ^* \underline{f}$$

$$\Rightarrow Q^* QR \underline{e} = Q^* \underline{f}_{||} \Rightarrow R \underline{e} = Q^* \underline{f}$$

$$\Rightarrow R \underline{e} = Q^* \underline{f}$$

$$Q^* \underline{f}_{||} = \overset{\sqrt{I}}{\cancel{Q^*}} (Q Q^* \underline{f}) = Q^* \underline{f} \checkmark$$

$\Rightarrow$ If E has lin. indep. columns, R is invertible.

Algorithm: Compute best approx. via QR

Step 1. QR decomposition of dictionary:

$$\begin{matrix} \begin{bmatrix} | & & | \\ \underline{e}_1 & \dots & \underline{e}_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ \underline{q}_1 & \dots & \underline{q}_n \\ | & & | \end{bmatrix} \begin{bmatrix} x & x & \dots & x \\ & x & & \vdots \\ & & \ddots & x \end{bmatrix} \\ E \qquad \qquad Q \qquad R \end{matrix}$$

Step 2. Solve upper triangular system.

$$R \underline{e} = Q^* \underline{f}$$

Question: What makes a good basis? "bad"

"good" $\bullet$ $\xrightarrow{\hspace{10em}}$ linear dependence

ONB

A good basis provides a stable representation of $f \in V$ under perturbations to coordinates of f .

Suppose $f = E \underline{c}$ and $\hat{f} = E \hat{\underline{c}}$, where

$$\|\underline{c} - \hat{\underline{c}}\| \leq \varepsilon \|\underline{c}\|$$

Q: How much bigger can $\|f - \hat{f}\|$ than $\varepsilon \|f\|$?

$$f - \hat{f} = E \underline{c} - E \hat{\underline{c}} = E(\underline{c} - \hat{\underline{c}})$$

$$\Rightarrow \|f - \hat{f}\| \leq \|E\| \|\underline{c} - \hat{\underline{c}}\| \leq \|E\| (\varepsilon \|\underline{c}\|)$$

The operator norm $\|E\| = \sup_{\underline{v} \in \mathbb{R}^n} \frac{\|E \underline{v}\|}{\|\underline{v}\|}$ measures the ability of $\underline{v} \mapsto E \underline{v}$ to amplify the size $\|\underline{v}\|$ to $\|E \underline{v}\|$. For ONB Q , the operator norm

$$\|Q\| = \sup_{\underline{v} \in \mathbb{R}^n} \frac{\|Q \underline{v}\|}{\|\underline{v}\|} = 1$$

$$\|Q \underline{v}\|^2 = \sum_{j=1}^n |\langle \underline{v}, \underline{q}_j \rangle|^2 = \sum_{j=1}^n |\underline{v}_j|^2 = \|\underline{v}\|^2$$

$$\Rightarrow \|f - \hat{f}\| \leq \varepsilon \|E\| \|c\|$$

$$\Rightarrow f = E c \Rightarrow f = (QR) c \Rightarrow Q^* f = R c$$

$$c = R^{-1} Q^* f$$

$$\|f - \hat{f}\| \leq \varepsilon \|E\| \|R^{-1} Q^* f\| \leq \varepsilon \|E\| \|R^{-1}\| \|Q^*\| \|f\|$$

$$\|f - \hat{f}\| \leq \varepsilon \overset{=1}{\cancel{\|Q\|}} \|R\| \|R^{-1}\| \overset{=1}{\cancel{\|Q^*\|}} \|f\|$$

$$\leq \varepsilon \kappa(R) \|f\| \quad \kappa(R) = \|R\| \|R^{-1}\|$$

"Condition #"

$\Rightarrow$ See demo 01.m on the course repo for an illustration of "good" vs. "bad" basis in the context of best L^2 -approximation by degree $\leq n$ polynomials.